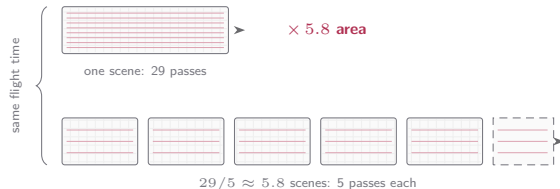
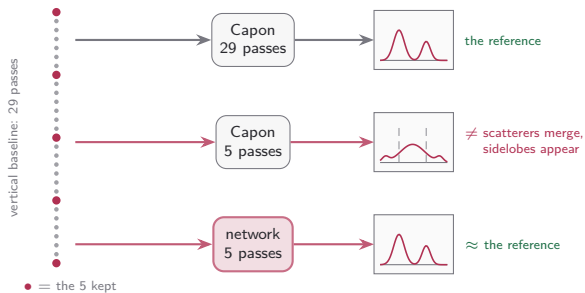


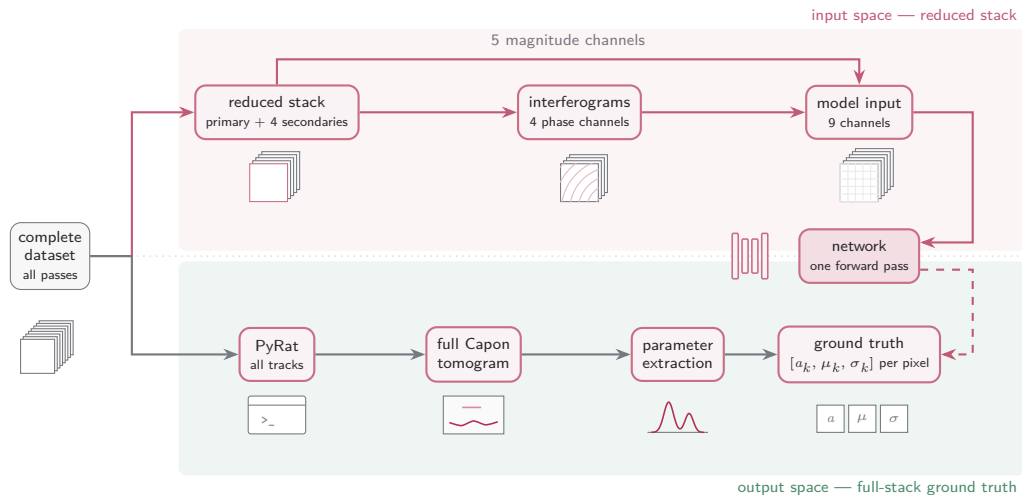
# Motivation: more area from the same flight time



Elevation resolution is bought with flight passes: every pass adds one vertical baseline sample over the scene.

- **The problem:** on the reduced stack the classical inversion degrades: with 5 baselines nearby scatterers merge and sidelobes contaminate the profile.
  - **The goal:** a network trained against full-stack ground truth recovers the full-stack parameters  $[a_k, \mu_k, \sigma_k]$  from the 5-pass stack alone.
  - **The payoff:** a scene then costs 5 passes instead of 29, so the same airtime surveys  $29/5 \approx 5.8$  scenes.
- **The question:** how close does the reconstruction come to the reference? The rest of the talk measures that.

# Methodology



A network learns to reconstruct, from the **reduced** stack alone, the per-pixel profile parameters that the full-stack tomogram and classical fit produce.

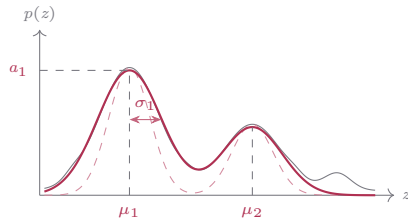
## Ground truth: fitted Gaussian parameters

Supervised target: the per-pixel array  $[a_k, \mu_k, \sigma_k]_{k=1}^K$  ( $K=2$  today,  $K=5$  in development), fit to the per-pixel max-normalised Capon spectrum  $\tilde{p} = p/p_{\max}$  on the elevation grid  $\xi_n$ :

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N (\hat{p}(\xi_n) - \tilde{p}(\xi_n))^2, \quad \hat{p}(\xi) = \sum_k a_k e^{-\frac{(\xi - \mu_k)^2}{2\sigma_k^2}}$$

One shared loss; each parameter group takes its own projected Adam step:

$$\begin{aligned} a_k &\leftarrow [a_k - \eta \text{Adam}(\partial_{a_k} \mathcal{L})]_+ \\ \mu_k &\leftarrow \text{clip}_{[\mu^-, \mu^+]} (\mu_k - \eta \text{Adam}(\partial_{\mu_k} \mathcal{L})) \\ \sigma_k &\leftarrow \text{clip}_{[\sigma^-, \sigma^+]} (\sigma_k - \eta \text{Adam}(\partial_{\sigma_k} \mathcal{L})) \end{aligned}$$

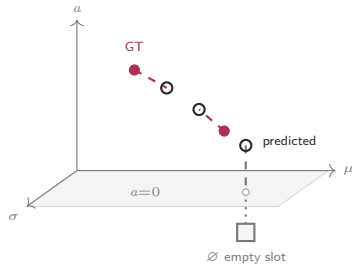


Same pixel: Capon spectrum, **peak initialisation (dashed)**, **converged fit**; the fitted  $[a_k, \mu_k, \sigma_k]$  are the label.

- Fitted in normalised space; the winning amplitudes are denormalised back,  $a_k \rightarrow p_{\max} a_k$  ( $\mu_k, \sigma_k$  already live on the physical axis).
- All pixels batched in one optimisation — not a per-pixel loop (next slide); most pixels activate one or two Gaussians (Act III).

# Hungarian matching

- Each Gaussian is a **point in  $(a, \mu, \sigma)$  space**: prediction and GT are two point sets; the matcher pairs them by distance and scores the **set**, not the slot layout.



Filled = active GT, open = predicted,  $\emptyset$  = empty GT slot (below  $a=0$ ).  
Its match (grey) is zero-weighted, so  $\hat{\theta}_3$  is free.

$$\mathcal{L} = \min_{\pi \in S_K} \sum_{k=1}^K w_k \left\| \hat{\theta}_{\pi(k)} - \theta_k \right\|^2 \quad \theta_k = (a_k, \mu_k, \sigma_k)$$

$w_k = 1[a_k > \tau]$  — GT *active mask*: 1 if present, 0 if empty.

|                       |                  | GT $\theta_j$ |            |             |
|-----------------------|------------------|---------------|------------|-------------|
|                       |                  | $\theta_1$    | $\theta_2$ | $\emptyset$ |
| pred $\hat{\theta}_i$ | $\hat{\theta}_1$ | 0.9           | 0.2        | 0.0         |
|                       | $\hat{\theta}_2$ | 0.4           | 0.8        | 0.0         |
|                       | $\hat{\theta}_3$ | 0.6           | 0.5        | 0.0         |

cost  $C_{ij} = w_j \|\hat{\theta}_i - \theta_j\|^2$

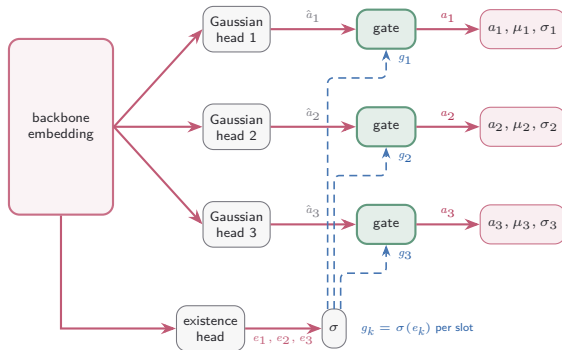
$\xrightarrow[\text{min cost}]{\pi^*}$

|                       |                  | $\theta_1$ | $\theta_2$ | $\emptyset$ |
|-----------------------|------------------|------------|------------|-------------|
| pred $\hat{\theta}_i$ | $\hat{\theta}_1$ | 0.9        | 0.2        | 0.0         |
|                       | $\hat{\theta}_2$ | 0.4        | 0.8        | 0.0         |
|                       | $\hat{\theta}_3$ | 0.6        | 0.5        | 0.0         |

$\sum C_{\pi^*} = 0.6$

Column  $\emptyset$  is a placeholder GT ( $w_j=0$ ) — all-zero, so the leftover prediction matches it for free:  $\hat{\theta}_1 \rightarrow \theta_2$ ,  $\hat{\theta}_2 \rightarrow \theta_1$ ,  $\hat{\theta}_3 \rightarrow \emptyset$ .

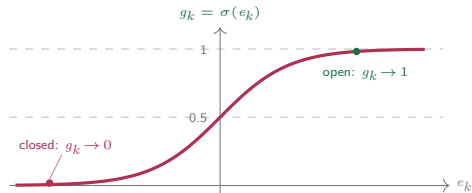
# Inside the set-prediction head — gate, blend, match



each gate:  $a_k = g_k \hat{a}_k + (1 - g_k) a_{\text{off},k}$   
 $\mu_k, \sigma_k$  pass through  $\cdot a_{\text{off}}$  learned per slot

The gate is a sigmoid on the existence logit  $e_k$ :

- Open slots ( $g_k \rightarrow 1$ ):  $a_k \approx \hat{a}_k$  — the raw regression survives.
- Closed slot ( $g_k \rightarrow 0$ ):  $a_k \rightarrow a_{\text{off},k} \approx 0$  whatever  $\hat{a}_k$  is, collapsing onto the empty-set target at near-zero loss.
- $e_k$  is internal — the head still emits  $3K$  channels ( $a_k, \mu_k, \sigma_k$ );  $a_{\text{off}}$  is a learned per-slot scalar (init 0).



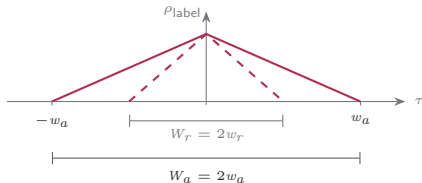
# Reading the reach — the network estimates statistics

two labels at offset  $\Delta$  split into shared and private looks:

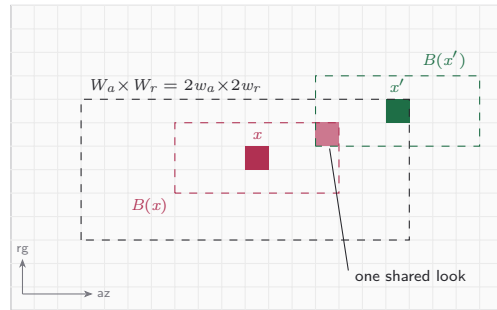
$$w_a w_r \hat{\mathbf{R}}(x) = \underbrace{\sum_{\mathcal{W} \cap \mathcal{W}'} y_p y_p^H}_{\text{shared } \mathbf{S}(\Delta)} + \underbrace{\sum_{\mathcal{W}' \setminus \mathcal{W}} y_p y_p^H}_{\text{private to } x}$$

$$w_a w_r \hat{\mathbf{R}}(x') = \underbrace{\sum_{\mathcal{W} \cap \mathcal{W}'} y_p y_p^H}_{\text{shared } \mathbf{S}(\Delta)} + \underbrace{\sum_{\mathcal{W}' \setminus \mathcal{W}} y_p y_p^H}_{\text{private to } x'}$$

the shared fraction at every offset  $\tau$  is the correlation law:



the label correlates over  $\pm w_a$  in azimuth and  $\pm w_r$  in range  
— the centred patch spanning both triangles is  $W_a \times W_r$ .



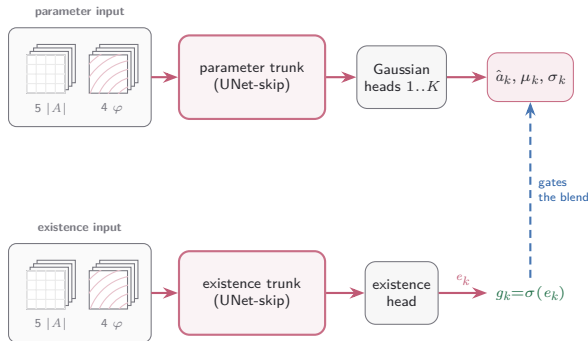
$x'$  sits at the far corner of the patch:  $B(x')$  keeps a single shared look with  $B(x)$  — one step further and the windows are disjoint ( $|\Delta_a| = w_a$  or  $|\Delta_r| = w_r$ ).

# The board at five seeds — the L1 family owns the top nine

| #  | model     | loss        | curve $R^2$ ↑      | cosine ↑           | SSIM<br>elev ↑ | SSIM<br>range ↑ | SSIM<br>azim ↑ | peak err ↓       | recall ↑    | 2-sc<br>recall ↑   | score        |
|----|-----------|-------------|--------------------|--------------------|----------------|-----------------|----------------|------------------|-------------|--------------------|--------------|
| 1  | unet-skip | L1-curve    | <b>0.756</b> ±.012 | 0.924 ±.007        | 0.9676 ±.0004  | 0.9500 ±.0016   | 0.9611 ±.0008  | 1.29 ±.05        | 0.886 ±.006 | 0.728 ±.013        | <b>0.983</b> |
| 2  | unet-skip | Charbonnier | 0.751 ±.012        | 0.928 ±.004        | 0.9675 ±.0007  | 0.9499 ±.0024   | 0.9608 ±.0012  | <b>1.25</b> ±.03 | 0.886 ±.004 | 0.727 ±.008        | 0.977        |
| 3  | unet-skip | param-L1    | 0.741 ±.016        | <b>0.931</b> ±.001 | 0.9669 ±.0004  | 0.9513 ±.0006   | 0.9608 ±.0005  | 1.38 ±.03        | 0.893 ±.004 | 0.745 ±.008        | 0.963        |
| 4  | resunet   | param-L1    | 0.734 ±.013        | 0.927 ±.007        | 0.9664 ±.0003  | 0.9502 ±.0006   | 0.9601 ±.0004  | 1.32 ±.07        | 0.894 ±.003 | <b>0.746</b> ±.006 | 0.946        |
| 5  | deeplab   | param-L1    | 0.731 ±.006        | 0.923 ±.004        | 0.9663 ±.0003  | 0.9495 ±.0004   | 0.9597 ±.0003  | 1.35 ±.05        | 0.888 ±.002 | 0.736 ±.005        | 0.937        |
| 6  | deeplab   | L1-curve    | 0.731 ±.012        | 0.921 ±.013        | 0.9667 ±.0005  | 0.9491 ±.0016   | 0.9599 ±.0008  | 1.33 ±.07        | 0.873 ±.022 | 0.721 ±.007        | 0.935        |
| 7  | resunet   | L1-curve    | 0.737 ±.019        | 0.916 ±.007        | 0.9668 ±.0004  | 0.9488 ±.0007   | 0.9601 ±.0006  | 1.34 ±.06        | 0.881 ±.005 | 0.715 ±.007        | 0.930        |
| 8  | resunet   | Charbonnier | 0.731 ±.016        | 0.921 ±.014        | 0.9668 ±.0006  | 0.9492 ±.0015   | 0.9601 ±.0006  | 1.31 ±.08        | 0.881 ±.006 | 0.716 ±.008        | 0.929        |
| 9  | deeplab   | Charbonnier | 0.735 ±.008        | 0.913 ±.015        | 0.9668 ±.0005  | 0.9492 ±.0020   | 0.9602 ±.0008  | 1.62 ±.45        | 0.868 ±.037 | 0.722 ±.013        | 0.912        |
| 10 | nafnet    | Charbonnier | 0.702 ±.009        | 0.906 ±.007        | 0.9645 ±.0004  | 0.9444 ±.0010   | 0.9569 ±.0004  | 1.48 ±.05        | 0.875 ±.008 | 0.711 ±.009        | 0.829        |
| 11 | segformer | Charbonnier | 0.688 ±.022        | 0.910 ±.016        | 0.9651 ±.0008  | 0.9447 ±.0028   | 0.9572 ±.0013  | 1.51 ±.20        | 0.866 ±.019 | 0.704 ±.019        | 0.829        |
| 12 | nafnet    | L1-curve    | 0.698 ±.007        | 0.908 ±.010        | 0.9646 ±.0005  | 0.9444 ±.0013   | 0.9568 ±.0007  | 1.43 ±.08        | 0.879 ±.005 | 0.714 ±.008        | 0.828        |
| ⋮  |           |             |                    |                    |                |                 |                |                  |             |                    |              |
| 16 | deeplab   | MSE-curve   | 0.732 ±.010        | 0.872 ±.019        | 0.9615 ±.0005  | 0.9368 ±.0011   | 0.9527 ±.0004  | 1.62 ±.08        | 0.869 ±.004 | 0.710 ±.003        | 0.785        |
| 20 | unet      | L1-curve    | 0.660 ±.015        | 0.897 ±.026        | 0.9649 ±.0009  | 0.9441 ±.0043   | 0.9563 ±.0018  | 1.51 ±.28        | 0.870 ±.029 | 0.711 ±.020        | 0.760        |
| 21 | unet-skip | param-MSE   | 0.647 ±.047        | 0.879 ±.017        | 0.9626 ±.0010  | 0.9398 ±.0032   | 0.9536 ±.0016  | 1.59 ±.13        | 0.881 ±.022 | 0.739 ±.014        | <b>0.692</b> |

runs: model · set-pred · Hungarian ·  $K=2$  · hv flips · presence A · loss · patch  $64 \times 32$ , five seeds; held-out band. Cells: mean ± seed std; **bold** = best mean in column (rows shown), marked only where best and worst differ by more than 5%. score = equal-weight mean of the five metric groups (curve error, variance, SSIM, cosine, peak), min-max scaled over the 35 arms; # = rank; ranks 13–15, 17–19, 22–35 on the next frame only. cosine = per-pixel profile cosine mean; peak err in elevation units; SSIM per axis vs the GT cube (denorm); recall = matched, 2-sc recall on GT  $k=2$  pixels. **green** = the winner's score; **red** = the param-MSE reference cell, 21st of 35.

# The dual model — two trunks, one head



$$\text{each gate: } a_k = g_k \hat{a}_k + (1 - g_k) a_{\text{off},k}$$

the set-prediction head of Act III, unchanged — only the source of each embedding splits

- **Split the encoder, keep the head:** two independent UNet-skip trunks — the Gaussian heads read one embedding, the existence head the other. Gate blend and Hungarian matching stay as they are.
- **Each trunk picks its own stack:** the channel groups  $|A|$  (5 amplitudes) and  $\varphi$  (4 interferograms) are selected per trunk — which modality reaches which head becomes an architectural choice.
- **Why route:** the input ablation split the labour — phase fires the slots, the full stack calibrates the parameters. The dual model can hand each head the input its job needs.



## Pairing the objective — param-L1 plus a curve term

$$\mathcal{L}(w) = (\mathcal{L}_{\text{param-L1}} + w \mathcal{L}_{\text{curve}}) / (1 + w)$$



1 baseline + 15 pair arms per  $K$ , five seeds each, at  $K=2$  and  $K=5$  = 160 runs

The ratio block left the  $K=5$  gate under-firing at every split — more existence capacity did not wake it. This experiment changes the objective instead: keep param-L1 and add a synthesized-curve term.

- **Two signals, one loss:** param-L1 reaches a slot only through its matched pair; the curve term re-synthesizes the profile from *all* slots and is permutation-invariant — its gradient touches every slot, matched or not.
- **Magnitude or shape:** L1 and Charbonnier price the synthesized profile's magnitude; cosine normalizes both curves first and prices shape alone.
- **A normalized pair:** the two terms are averaged, so the validation loss changes meaning with  $w$  — arms compare on held-out metrics only, never on the loss.
- **The question:** if the conservative gate is starved of a profile-level error signal rather than capacity, a small  $w$  should move detection at  $K=5$  and leave  $K=2$  alone.

## Next steps

- **Physical loss terms:** the five ranked terms on the next three frames: covariance matching, coherence re-synthesis, moments, total power, the Capon cycle.
- **$\gamma$ -net adaptation:** adapting the  $\gamma$ -net architecture to this pipeline.
- **JEPA pretraining:** predicting in representation space, the closing frame.

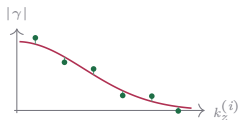
# Data-consistency terms (the ranked top two)

## Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n) e^{jk_z^{(i)} \xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i |\gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)})|^2 \right\rangle$$



$\gamma_P$  from the predicted mixture pulled onto the target coherences,  
baseline by baseline.

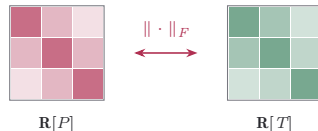
Cloude 2006 (PCT); same Fourier kernel as gamma-Net  
(Qian et al.).

## Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A} \text{diag}(P) \mathbf{A}^H \Delta \xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale,  
entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros &  
Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

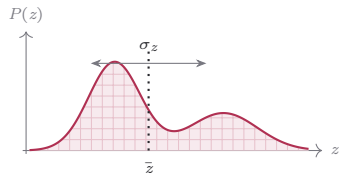
# Descriptor terms: moments and total power

## Moments

mass · phase-centre  $\bar{z}$  · spread  $\sigma_z$

$$m = \sum_n P(\xi_n) \Delta \xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \quad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



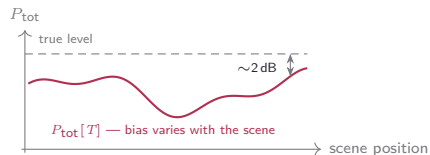
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

## Total power

the one absolute-power quantity

$$P_{\text{tot}}[\cdot] = \sum_n (\cdot)(\xi_n) \Delta \xi$$

$$\ell_{\text{pow}} = \left\langle (P_{\text{tot}}[P] - P_{\text{tot}}[T])^2 \right\rangle$$

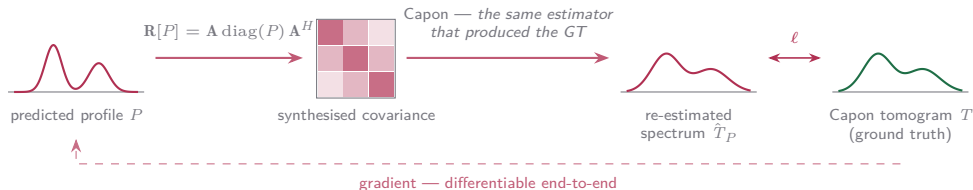


The target  $P_{\text{tot}}[T]$  is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

# The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

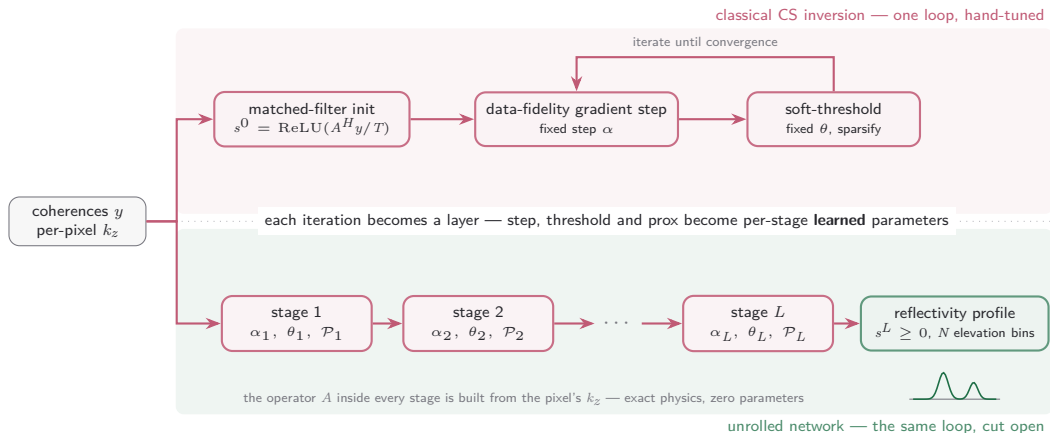
$$\hat{T}_P(\xi_n) = (\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n))^{-1}$$



- Absorbs the Capon bias and the PSF **by construction**.
- Costs an  $N_s \times N_s$  solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

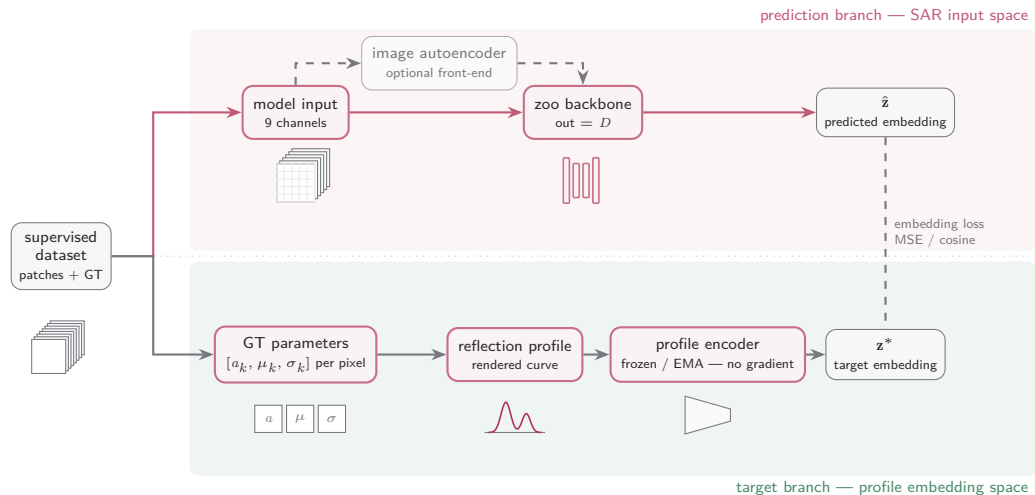
# The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \operatorname{Re}[A^H(y - As^\ell)]/L_{\text{lip}}, \quad s^{\ell+1} = \operatorname{ReLU}(\mathcal{P}_\ell(r^\ell) - \theta_\ell)$$

$\gamma$ -Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018).  $\mathcal{P}_\ell$ : per-pixel residual 1D convolution along elevation;  $L_{\text{lip}} = TN \mathrm{d}z^2$  normalises the step. Default  $L=8$ :  $\sim 1.4\text{k}$  parameters.

# What JEPA does



A network learns to predict, from the reduced stack alone, the **embedding** that a separately-trained profile autoencoder assigns to the ground-truth reflection profile.